
Force-distilled invariant descriptors: A four-metal test of quadratic potential learning

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Abstract

Learning an atomic descriptor should be evaluated separately from changing the energy regression. We introduce force-distilled radial modes (FDRM), a supervised spectral compression of Cartesian atomic moments, followed by a separately fitted quadratic least-squares head. The 40-channel descriptor preserves a fitted linear pilot exactly, enforces rotation, translation, and permutation invariance, and uses 860 head coefficients instead of quadratic SNAP’s 1595. Complete five-percentage-point learning curves with ten draws per fraction cover tantalum and three independently refitted refractory elements. The anticipated efficiency gain does not materialize. On Ta, pooled full-data energy/force errors are 0.000353 eV/atom and 0.02784 eV/Å, versus 0.000242 and 0.01611 for refitted SNAP; neither the twelve-group aggregate nor the low-symmetry subset reaches SNAP’s full-data energy threshold by 95% included data. Mo exhibits only a late, nonmonotone crossing, while Nb and Re regress. Symmetry drift remains tiny, and fixed-mode, random-mode, short-range-reference, and duplicate-blocked controls clarify the failure. Compressed crystals with almost zero atomic forces can dominate excluded energy error, whereas liquid force errors expose lost representational capacity. Exact symmetry and retention of a supervised pilot are therefore insufficient certificates of sample efficiency for this quadratic extension.

1 What should learning a descriptor buy?

A quadratic atomic energy model can improve markedly on a linear one while retaining a linear fitting problem. Wood and Thompson’s quadratic SNAP construction makes this tradeoff explicit: the bispectrum is fixed, and accuracy is purchased with products of its components [Wood and Thompson, 2018]. Replacing that bispectrum with a learned representation raises a more specific question than whether neural potentials can be accurate. Can a descriptor adapted to the energy and force response reduce the data required by the *same quadratic regression form*?

We study this question with *force-distilled radial modes* (FDRM). A linear pilot potential learns which radial combinations of Cartesian atomic moments contribute to the observed response. Diagonalizing its coefficient matrices produces a compact, invariant descriptor; a fresh linear-plus-quadratic head is then fit by least squares. Descriptor learning is therefore distinct from fitting the final energy model. This construction permits a useful identity: the compact descriptor exactly represents the pilot potential, even though it discards most independent radial cross-contractions. The quadratic stage can improve the pilot’s training objective, but this identity supplies no guarantee about excluded configurations.

The experiment is designed to expose that distinction. We evaluate every five-percentage-point training fraction with ten configuration-level draws on Ta, Mo, Nb, and Re, retaining a separately fitted quadratic SNAP comparator. We report both configuration-pooled and equally group-weighted errors, and isolate Ta’s stacking-fault, liquid, and surface groups. Symmetry transformations, fixed and random radial-mode controls, a short-range-reference sensitivity study, and a duplicate-blocked split control distinguish representation, coverage, and implementation effects. The resulting evidence

supports a negative answer for this particular compression rule: exact invariance and a smaller quadratic head do not establish accuracy parity or a half-data advantage. The failures themselves reveal why a force-informed pilot can leave large energy-curvature errors unconstrained in highly symmetric, compressed crystals.

Relation to learned atomic representations. Moment tensor potentials and the atomic cluster expansion establish the invariant contraction algebra used here [Shapeev, 2016, Drautz, 2019]. Proper orthogonal descriptors compress a parameterized potential family through a Karhunen–Loève construction [Nguyen and Rohskopf, 2023]; tensor-reduced density representations factorize density products and address their scaling, especially with chemical diversity [Darby et al., 2023]. FDRM instead diagonalizes matrices *fitted to energy and force labels*, independently in each training draw. Its squared radial-mode norms are familiar invariant features, not a new invariant basis. The contribution is the response-based training rule and its controlled evaluation with a fixed quadratic energy head.

CACE combines Cartesian contractions with trainable channel coupling, chemical embeddings, and message passing [Cheng, 2024]. NequIP and MACE learn equivariant neural potentials and demonstrate that architectural symmetry can coexist with data efficiency [Batzner et al., 2022, Batatia et al., 2022]. FDRM has neither message passing nor a neural energy network, and does not reproduce those architectures or their training procedures. Their successful results motivate the question rather than predict its answer for the more restricted construction considered here.

2 Distilling a response into invariant radial modes

All architectural and fitting choices in this section are shared across materials. Only the fitted coefficients and mode matrices change. No weights are transferred between elements, and no configurations are augmented by symmetry transformations during training.

2.1 Geometry, moments, and a supervised spectral map

For the periodic neighbors of atom i , write $r_{ij} = \|\mathbf{r}_{ij}\|$ and $\mathbf{u}_{ij} = \mathbf{r}_{ij}/r_{ij}$. Periodic images are included within $r_c = 5.594$ Å. Eight radial functions are fixed before fitting:

$$g_n(r) = \exp[-((r - c_n)/s)^2]f_c(r), \quad f_c(r) = \frac{1}{2}[1 + \cos(\pi r/r_c)], \quad 0 \leq r < r_c, \quad (1)$$

$$s = 0.65 \text{ Å}, \quad c_n = 1.5 + (n - 1)(r_c - 1.8)/7 \text{ Å}. \quad (2)$$

The functions and their first derivatives vanish at the cutoff; $g_n = 0$ outside it. Define scalar densities $d_n^i = \sum_j g_n(r_{ij})$ and Cartesian tensors

$$M_{n\ell}^i = \sum_j g_n(r_{ij}) \mathbf{u}_{ij}^{\otimes \ell}, \quad P_{nm\ell}^i = M_{n\ell}^i : M_{m\ell}^i, \quad \ell = 1, 2, 3, 4. \quad (3)$$

The colon denotes complete contraction over the ℓ Cartesian indices. The implementation retains all 3^ℓ components; it does not use an irreducible harmonic decomposition. Each contraction is a sum of products weighted by $(\mathbf{u}_{ij} \cdot \mathbf{u}_{ik})^\ell$. The pilot basis has $8 + 4 \times 36 = 152$ scalar components, using $n \leq m$.

Fit a linear pilot energy on the included configurations only:

$$E_{\text{pilot}} = \sum_i \left[\mathbf{a} \cdot \mathbf{d}^i + \sum_{\ell=1}^4 \sum_{n,m=1}^8 A_{nm\ell} P_{nm\ell}^i \right]. \quad (4)$$

Each A_ℓ is symmetric; an off-diagonal coefficient of the upper-triangular fitting basis is divided between its two symmetric entries. Let $A_\ell = U_\ell \Lambda_\ell U_\ell^\top$, with eigenvalues ordered by decreasing absolute magnitude. The learned descriptor is

$$\mathbf{B}^i = (\mathbf{d}^i, \{b_{k\ell}^i\}_{k=1,\dots,8;\ell=1,\dots,4}), \quad b_{k\ell}^i = \left\| \sum_n U_{nk\ell} M_{n\ell}^i \right\|_F^2. \quad (5)$$

All eight modes of each matrix are retained. Thus compression acts on the 36 independent cross-contractions per order, rather than by truncating the radial eigenbasis: the descriptor has $K' = 40$

components. Its four learned orthogonal matrices have 256 entries and 112 continuous degrees of freedom; prediction artifacts store their expanded 152×40 feature map. Signs of eigenvectors are immaterial because their tensor combinations are squared. Near-degenerate eigenspaces can nevertheless rotate between draws, changing the retained squared modes and the ensuing quadratic hypothesis class.

2.2 A fixed quadratic head and what the compression preserves

Freeze the mode matrices and fit 40 linear and 820 upper-triangular quadratic coefficients:

$$E = \sum_i \left[\beta^\top \mathbf{B}^i + \frac{1}{2} (\mathbf{B}^i)^\top \alpha \mathbf{B}^i \right], \quad \mathbf{F}_j = -\nabla_{\mathbf{r}_j} E. \quad (6)$$

The symmetric matrix convention absorbs the diagonal factor of two in an upper-triangular product implementation. There is no free constant. Every channel vanishes for an atom without neighbors, so its predicted energy is exactly zero. All force rows are analytic derivatives of Eq. (6); the efficient evaluator contracts the energy gradient back through the radial and tensor operations, without fitting a separate force function.

Because $\sum_{nm\ell} A_{nm\ell} P_{nm\ell}^i = \sum_k \Lambda_{kk\ell} b_{k\ell}^i$, setting the quadratic head to zero and its linear coefficients to the pilot coefficients recovers Eq. (4) for *every* configuration. An exact least-squares fit of the larger head cannot increase the included weighted squared error. It can still amplify directions weakly constrained by the training set. Moreover, each $b_{k\ell}$ is at most a three-body feature; multiplying two such features gives at most five-body interactions. Increasing tensor order in Eq. (3) increases angular resolution, not body order. Quadratic SNAP’s bispectrum products can represent seven-body terms [Wood and Thompson, 2018]. Matching the algebraic head therefore does not equalize representational capacity.

For both the pilot and final head we minimize

$$\sum_{c \in \mathcal{T}} \left[10^4 (\hat{E}_c / N_c - E_c / N_c)^2 + \sum_{j,a} (\hat{F}_{cja} - F_{cja})^2 \right]. \quad (7)$$

Columns are divided by their weighted included-data Euclidean norms, with a 10^{-14} floor, and the minimum-norm scaled solution uses a relative singular-value cutoff of 10^{-13} . No ridge penalty, early stopping, hyperparameter search, or excluded-set model selection is used. The solve is a single SVD, not an iterative neural training procedure. A GPU QR reduction precedes the unchanged FitSNAP SVD solver for the sweep; it avoids normal equations. Geometry and its derivatives may be cached globally, but every pilot, mode matrix, column scale, and head is fitted anew using only $\mathcal{T}$. Fixed-mode and random-mode controls replace each U_ℓ by the identity or a seeded random orthogonal matrix, respectively, while preserving the 40-channel architecture and final fitting rule.

2.3 Invariance follows from the representation

Translation leaves each neighbor vector unchanged. A rotation R acts on a moment as $R^{\otimes \ell} M_{n\ell}$; orthogonality makes its contracted norm invariant. Neighbor summation makes the descriptor invariant to permutations of like atoms. Learned radial mixtures commute with the rotation, so neither label fitting nor its eigenvectors can break these symmetries. Differentiating the invariant energy yields rotation-equivariant and permutation-equivariant forces. The construction is also reflection invariant. The empirical audit below tests floating-point realizations of these statements, including transformations of the periodic cell.

3 Data, paired splits, and numerical accountability

Data provenance. The Ta study retains all 363 JSON configurations in the archived FitSNAP Ta example, organized in twelve groups. Direct enumeration gives 4224 atoms, 12672 force components, and 2–100 atoms per configuration. These files differ from the source paper’s stated 362 energy and 15297 force observations; we neither create a missing observation nor remove an available one. The three transfer collections are the supplied Mo, Nb, and Re archives from Andolina and Saidi [2023]. We convert their DeePMD energy, position, cell, and force arrays to individual FitSNAP

JSON records without label filtering, offsets, or rescaling. The resulting counts are 3727, 3246, and 5420 configurations, respectively. Their atom-count ranges are 7–32, 1–32, and 7–32. A one-atom *periodic* Nb cell is retained as a crystal, not treated as an isolated atom.

Transfer strata are archive iteration crossed with the source directory’s elastic, vacancy, or remaining trajectory category, yielding nine Mo, twelve Nb, and nine Re groups. Temperatures and crystal prototypes are pooled within these strata; the original per-configuration archive paths remain in the logs. The transfer experiment tests reuse of an architecture with new fitted weights, not zero-shot chemical transfer. No OMatG-generated structures, external Ta configurations, or new DFT calculations enter fitting.

Splits and metrics. At each nominal fraction $f = 0.05, 0.10, \dots, 0.95$, ten independently seeded draws select configurations within every stratum. The included count in a group of size n is $\lfloor fn + 0.5 \rfloor$, clipped to $[1, n - 1]$; all energies and force components of a configuration stay together. Seeds are $20261004 + 1000(100f) + d$, for draw $d \in \{0, \dots, 9\}$. The paired methods receive identical index lists. The 100% fit is performed once and has no excluded observations. Means and sample standard deviations use all ten draws; uncertainty bands are standard deviations, not confidence intervals.

Finite group sizes matter. The nominal Ta budgets 5%, 50%, and 95% actually include 22, 185, and 342 configurations. Liquid contains only three configurations: its split can use only one or two before the 100% endpoint. The endpoint is always identified as a *fitting error*, never a held-out accuracy measurement.

Energy MAE averages $|\hat{E}_c/N_c - E_c/N_c|$ over configurations. Force MAE averages absolute Cartesian-component errors. “Pooled” means all configurations or all force components; “equal group” first computes each group’s MAE, then averages groups equally. Ta’s low-symmetry aggregate equally weights GSF_110, GSF_112, Liquid, and Surface. For each material we define F^* as the first measured excluded-error mean no larger than that material’s quadratic SNAP 100% fitting energy MAE under the same aggregation. We additionally report the first crossing sustained at every later measured fraction. The reference budget is always 100%; absence of a crossing is a negative result.

Baselines and departures from the published fit. The external anchor is quadratic SNAP at $J_{\max} = 4$, $r_c = 5.594 \text{ \AA}$, and $\delta E = 0.00898 \text{ eV/atom}$ (Wood and Thompson, Supplemental Table SII). The paired refit uses the bundled FitSNAP calculator (commit 1a9b050) with `twojmax=8`, `rfac0=0.99363`, 55 bispectrum components, 1595 fitted coefficients, and `bzeroflag=1`. We hold the cutoff at $5.594 \text{ \AA}$ for every material rather than independently optimizing it for Mo, Nb, and Re. This deliberately narrow test does not estimate either method’s best achievable transfer-material accuracy.

The main comparison uses a zero reference pair potential, energy/force weights 100/1 in every group, and no stress observations. It therefore differs from the supplied quadratic example’s ZBL overlay and negligible stress weights, and from the source paper’s optimized group-weight workflow. Both learned and SNAP models share these choices. The zero isolated-atom energy, conservative forces, complete split sweep, quadratic head, and original supplied configurations are preserved. To assess the reference choice, a separate Ta control restores the supplied ZBL reference (switching between 4.0 and 4.8 $\text{\AA}$; atomic number 73) for both methods at 20%, 50%, 80%, 95%, and 100%, with ten draws below 100%. We also evaluate the supplied 2021 FitSNAP coefficient artifact directly. These controls remain separate from the 2018 published number and the main paired refits.

Audits and cost. For each material, ten configurations are randomly selected from the excluded portion of the 80%, draw-0 split. Each receives 100 Haar-random rotations of coordinates and cell, ten translations with components uniform on $[-1, 1] \text{ \AA}$, and ten permutations. Predictions, transformations, and untransformed references are stored. Forces are rotated back or inverse-permuted before comparison. We report absolute maximum force drift as well as energy-relative and component-relative drift, with denominator floor 10^{-12} for the latter two. This explicitly handles nearly force-free crystals.

Inference timing uses 128-atom periodic systems, BCC for Ta/Mo/Nb and HCP for Re, initialized at 300 K. Single-thread CPU velocity-Verlet trajectories use a 0.2 fs timestep, five warm-up steps, and seven blocks of ten measured steps. SNAP runs through the exported LAMMPS pair potential [Thompson et al., 2022]; FDRM uses an analytic NumPy/ASE evaluator. Seven energy-only and

Material	Method	Configuration pooled		Equal group		F^*	MD rel.
		δE	δF	δE	δF		
Ta	SNAP (2018) [†]	0.00898	—	—	—	—	—
Ta	SNAP (refit)	0.0002424	0.01611	0.0003204	0.01218	reference	1.00
Ta	FDRM	0.0003531	0.02784	0.0004932	0.02309	NR	2.9
Mo	SNAP (refit)	0.02591	0.483	0.05831	0.6255	reference	1.00
Mo	FDRM	0.0293	0.5222	0.06176	0.6077	85%	3.3
Nb	SNAP (refit)	0.004898	0.1369	0.003949	0.1254	reference	1.00
Nb	FDRM	0.01377	0.5383	0.0129	0.4824	NR	2.9
Re	SNAP (refit)	0.007444	0.3172	0.008912	0.32	reference	1.00
Re	FDRM	0.03529	1.174	0.04398	1.222	NR	3.6

Table 1: Full-data fitting errors and measured inference cost. Energies are in eV/atom and forces in eV/Å. Pooled force errors weight components equally; equal-group errors weight strata equally. F^* uses excluded equal-group energy means and the refitted SNAP full-data reference. NR means no crossing at any measured 5–95% fraction. SNAP’s data budget and timing are the reference in each material. [†]The 2018 value is exactly Supplemental Table SII, $J_{\max} = 4$, under that paper’s reporting convention; no published force value is inferred. The MD ratio compares implementations with different observed accuracies, not a matched-accuracy speedup.

energy-plus-force timings also isolate evaluation overhead. These short trajectories are cost probes, not tests of thermodynamic validity or long-time stability. Fits use one NVIDIA H200 and up to four BLAS threads; CPU timing uses an Intel Xeon Platinum 8558. The observed hardware is reported rather than substituting a nominal accelerator specification.

4 Accuracy, sample budgets, and structural failures

4.1 A compact invariant descriptor does not establish accuracy parity

Table 1 separates the published anchor from our measured refits. On Ta, FDRM’s configuration-pooled full-data energy and force MAEs are 0.000353121 eV/atom and 0.0278439 eV/Å, versus 0.000242415 and 0.0161124 for quadratic SNAP. Equal-group averaging leads to the same ordering. Consequently this descriptor does *not* meet simultaneous energy-and-force accuracy parity. This is a statement about these completed fits, not a prediction about an unrun optimization.

The transfer results also resist a single favorable summary. FDRM’s full-data equal-group energy errors exceed SNAP’s on all three materials. On Mo its equal-group force error is slightly lower, 0.607694 versus 0.625484 eV/Å, but pooling force components reverses that comparison: 0.522219 versus 0.482969. On Nb and Re both aggregations favor SNAP decisively. Re’s equal-group force MAE rises from 0.32001 to 1.22232 eV/Å. This HCP-ground-state element is a useful breadth test, but chemistry, structure populations, and reference-label distributions change together; the result does not identify HCP symmetry as the cause.

4.2 The complete learning curves reject a half-data claim

For neither the equal-weight twelve-group aggregate nor the low-symmetry subset does FDRM reach the corresponding full-data SNAP energy error at any measured excluded-data fraction. Thus F^* is not reached in either Ta view, and the stronger “at most half the data” claim is unsupported. At 50%, the all-group energy mean is 23.54 ± 27.75 eV/atom, while the low-symmetry mean is 0.00837 ± 0.00555 eV/atom. Their reference thresholds are 0.0003204 and 0.0005561 eV/atom, respectively. Reporting just the favorable subset comparison against SNAP at the *same* 50% budget (0.01630 ± 0.01721 eV/atom) would answer a different question from matching its full-data error.

Figure 2 explains the enormous aggregate/subset separation. The volume scans dominate the unstable tails. In Volume_BCC, excluded energy MAE at 50% is 256.5 ± 331.4 eV/atom for FDRM and 103.0 ± 106.8 for SNAP. These errors are neither removed nor winsorized. Ten independent draws need not yield monotone curves: at 95%, withholding a rare, highly compressed configuration can still cause a large error. Conversely, the tiny training errors at 100% do not establish reliable interpolation between volume points.

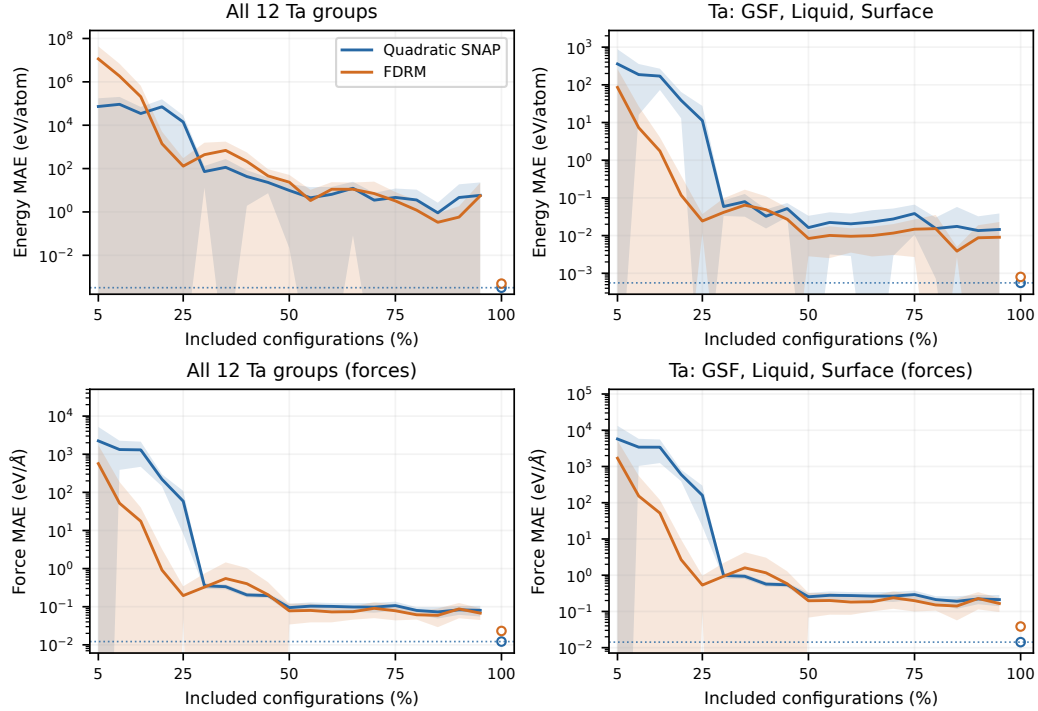


Figure 1: Ta learning curves, using equal group weights for all twelve groups (left) and for GSF_110, GSF_112, Liquid, and Surface (right). Curves and bands show the excluded mean and one sample standard deviation over all ten draws. Hollow circles at 100% are fitting errors; they are deliberately disconnected from the held-out curves. Dotted lines mark refitted SNAP’s full-data fitting error. Lower bands are clipped to a positive display floor on logarithmic axes; exact means and standard deviations are available in the accompanying CSV.

The transfer curves in Fig. 3 are numerically smoother but give no half-data result. Mo first crosses its full-data SNAP energy threshold at 85%, corresponding to only $1/0.85 = 1.18$ times fewer configurations by the nominal budget. That crossing is lost at 90%; the sustained crossing is 95%, a 1.05-fold reduction. Neither Nb nor Re crosses by 95%. At 50%, FDRM’s equal-group energy MAEs are 0.06864 ± 0.00729 , 0.01680 ± 0.00421 , and 0.04313 ± 0.01664 eV/atom for Mo, Nb, and Re. The same 100%-budget reference definition is used throughout; no material is assigned an easier reference fraction.

4.3 Where the representation loses information

Learning the modes is useful in some comparisons, but not sufficient. With the same 860-coefficient head, full-data equal-group force MAEs on Re are 1.222, 1.415, and 1.381 eV/Å for supervised, fixed, and random modes. On Mo the corresponding errors are 0.608, 0.713, and 0.652. On Ta, however, they are nearly indistinguishable: 0.02309, 0.02319, and 0.02297. The fixed and random controls are evaluated at 20%, 50%, and 80% with ten draws and at the full-data endpoint; Table 4 reports the 80% comparisons. There is no consistent benefit from the supervised orientation across materials and metrics.

The algebra explains why good pilot retention can coexist with this outcome. Distillation preserves a *single fitted linear response*, not all independent cross-contractions and not all higher-body correlations. The quadratic fit can couple retained squared modes but cannot restore every discarded direction. At full-data Ta the measured weighted squared residual decreases from 293.38 for the pilot to 35.66 for FDRM, confirming the nesting argument; improved included loss plainly does not imply a successful excluded curve. Larger angular tensor order cannot compensate automatically for the

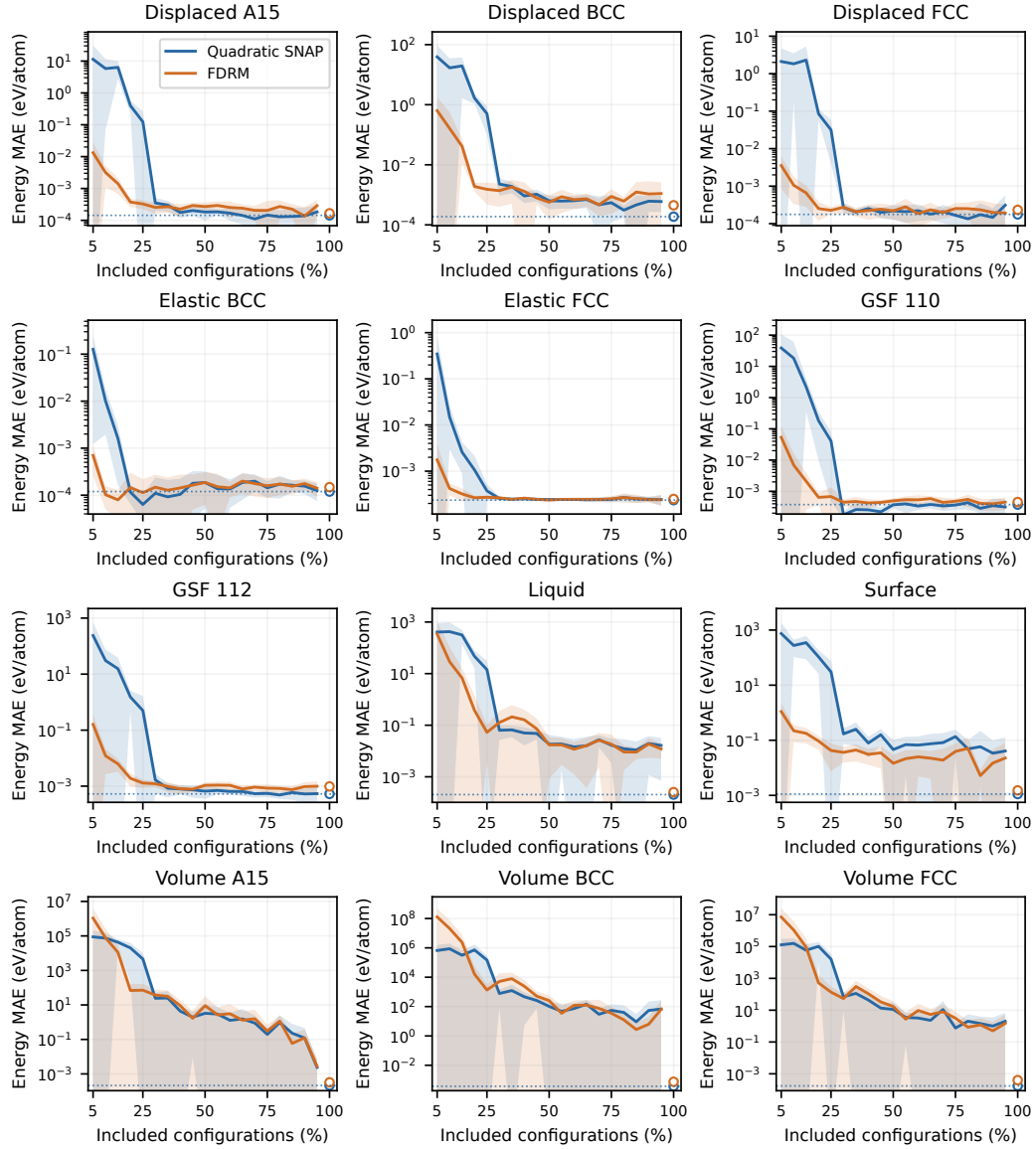


Figure 2: All twelve Ta groups, with no group privileged or omitted. The conventions are those of Fig. 1. The contrast between force-free volume scans, displaced crystals, and small low-symmetry groups is central to interpreting sample efficiency. Different vertical scales retain visibility of both small fitting errors and catastrophic excluded errors.

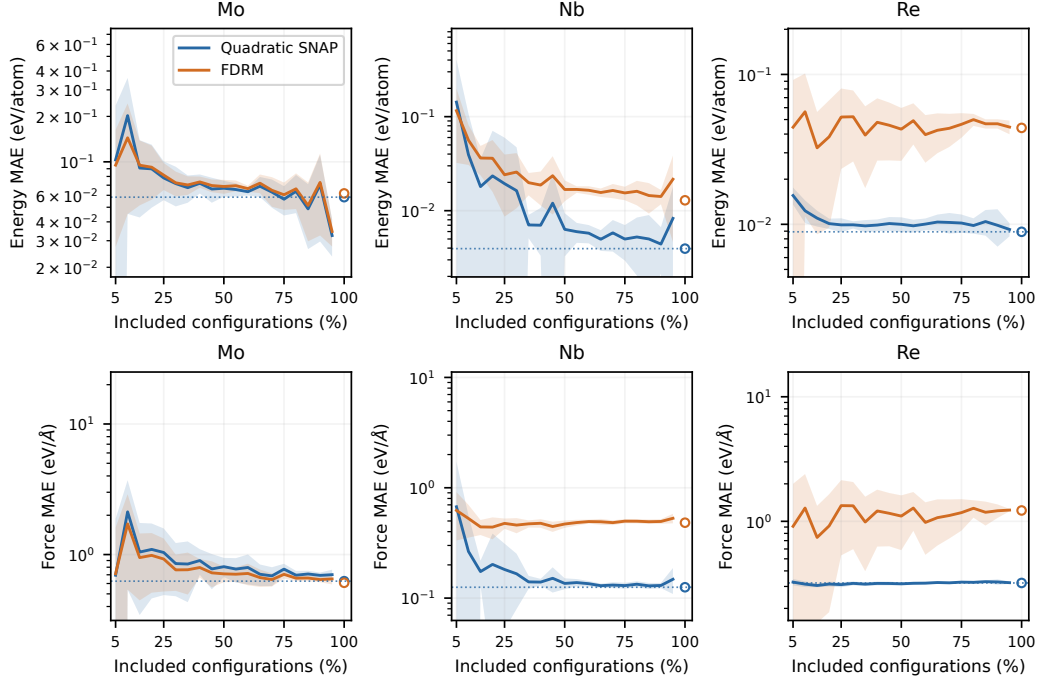


Figure 3: Complete transfer-material sweeps, retraining the same architecture separately on each element. Each point averages the original iteration/perturbation strata equally; bands include all ten draws. Energy and force curves are reported together. Full-data fitting points and SNAP reference lines follow Fig. 1. Per-stratum curves appear in the appendix, and every material–fraction–draw–group cell is retained in the logs.

difference in maximum body order from quadratic SNAP. These are plausible capacity mechanisms; without a separately controlled higher-body extension, the experiments do not isolate a unique cause.

The largest Ta energy failures have a more direct structural interpretation. Perfect volume-scan crystals have nearly vanishing atomic forces even when their hydrostatic energy slope is large. With stress omitted, their force rows supply essentially no information about that slope. In the 80%, draw-0 split, the two-atom BCC cell at $6.0835 \text{ \AA}^3/\text{atom}$ has an energy error of 168.44 eV/atom for SNAP and 34.03 for FDRM, while force errors remain of order 10^{-12} and $10^{-11} \text{ eV/\AA}$. High symmetry has removed force information; it has not guaranteed coverage of the equation of state. Liquid presents the complementary failure: at full data, Ta liquid force MAE is 0.10634 for FDRM versus 0.02605 eV/Å for SNAP, despite small energy fitting errors. The two failures should not be collapsed into a generic claim of insufficient training data.

4.4 Published-reference and duplicate-split checks

Our pooled Ta SNAP full-data energy MAE, 0.0002424 eV/atom, is 0.0087376 eV/atom below the 2018 Table SII value, a factor of 37.04. This substantial disagreement must constrain interpretation: the small refit error is *not* a reproduction of the published numerical endpoint or evidence of improved literature-level generalization. The supplied file inventory, fitting weights, reference potential, and scaled numerical solve differ from the published workflow.

Two measured checks narrow the explanation. Evaluating the supplied 2021 coefficient artifact with its ZBL overlay gives 0.000238497 eV/atom and 0.0158500 eV/Å pooled errors, reproducing its accompanying reference metrics. The data/calculator path therefore agrees with that supplied artifact. Restoring ZBL and refitting our Ta comparison yields equal-group full-data energy errors 0.0003143 for SNAP and 0.0004944 for FDRM; the learned model still lacks parity. The excluded volume errors are sensitive to this prior: at 95%, the SNAP all-group mean falls from 5.805 to 0.455 eV/atom, whereas FDRM changes from 5.538 to 6.138. At 50%, their low-symmetry errors are still 0.01965

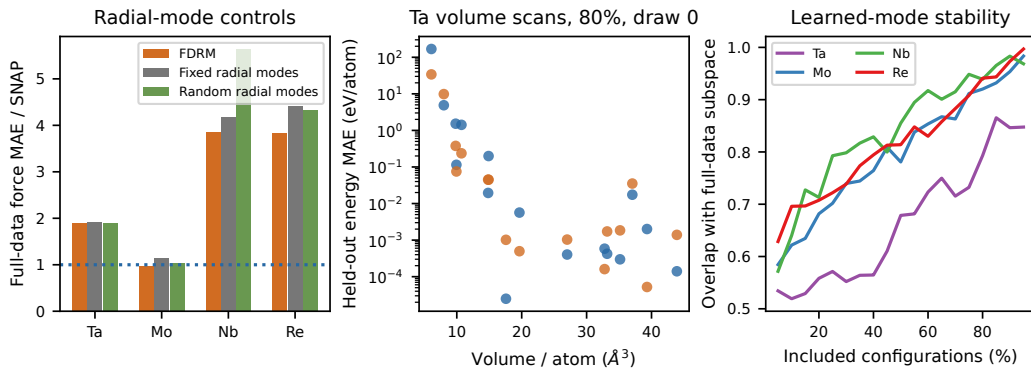


Figure 4: Mechanism checks. Left: full-data equal-group force errors relative to SNAP, contrasting supervised, fixed-axis, and random radial modes at identical descriptor and head dimensions. Center: errors for excluded Ta volume configurations at the fixed 80%, draw-0 split, against their volume per atom. Right: mean squared subspace overlap with each material’s full-data learned map, measured in the fixed raw coefficient basis. This retrospective stability diagnostic is never used to choose a model or tune a split.

and 0.00944 eV/atom, far above their full-data reference. The failure of the half-data claim persists, but the size of the most severe tail depends on the reference choice. The remaining disagreement with 2018 cannot be assigned to ZBL alone; we do not claim to have recovered the publication’s complete optimization history.

Coordinate hashes also identify 64, 117, and 380 duplicate-coordinate clusters in Mo, Nb, and Re, respectively, and none in Ta. The random-configuration curves consequently need not estimate generalization to genuinely new geometries. A separate ten-draw control selects 80% of whole coordinate clusters, without source-stratum balancing, and never splits a cluster. Its mean equal-group energy MAEs for SNAP/FDRM are 0.06474/0.06827 on Mo, 0.00652/0.01778 on Nb, and 0.00979/0.04884 eV/atom on Re. The ordering remains unfavorable to FDRM. This control removes exact duplicates at the stated hash precision, not temporal correlations or symmetry-equivalent structures, and its different stratification prevents attributing every numerical change solely to duplication.

5 Symmetry and implementation cost

The invariance audit supports the architectural argument while showing the hazards of relative force errors (Table 2). Across all tests, FDRM’s maximum energy-relative drift is 5.87×10^{-13} and absolute force drift is 2.41×10^{-11} eV/ $\text{\AA}$. The corresponding SNAP maxima are 1.80×10^{-13} and 4.25×10^{-12} . Accumulation and cancellation in fitted floating-point expressions amplify machine epsilon; neither method literally returns bitwise-identical predictions after all transformations.

The largest Ta component-relative force drifts are 5.59 for FDRM and 4.25 for SNAP. They do not indicate forces changing by several physically meaningful units: the denominators are approximately the 10^{-12} eV/ $\text{\AA}$ numerical floor in nearly force-free volume configurations. On the other three materials, maximum component-relative drifts are at most 1.02×10^{-9} for FDRM and 2.71×10^{-11} for SNAP. These results tie the apparent relative failure to the chosen normalization. The observed predictive force errors, up to order-one eV/ $\text{\AA}$, exceed absolute transformation drifts by many orders of magnitude. Broken rotation, translation, or permutation symmetry is therefore not a viable explanation for the accuracy deficit. The audit used no augmented training data.

Table 3 distinguishes fitting convenience from deployable inference speed. One-time feature extraction sums to 709 s for SNAP and 735 s for the moment basis across all materials (summed configuration wall times, with transfer work distributed over eight workers). The cached 40-channel regression is inexpensive, but the present NumPy evaluator explicitly constructs Cartesian tensor components and radial contractions and rebuilds its neighbor list at each call. LAMMPS supplies

Material	Method	Test	$\max \Delta E / E $	$\max \Delta F $	$\max d_{F,\text{comp}}$
Ta	SNAP	rotation	1.8×10^{-13}	4.25×10^{-12}	4.25
Ta	SNAP	translation	1.48×10^{-13}	2.69×10^{-12}	2.69
Ta	SNAP	permutation	8.45×10^{-14}	1.46×10^{-12}	1.24
Ta	FDRM	rotation	5.87×10^{-13}	2.41×10^{-11}	5.59
Ta	FDRM	translation	3.21×10^{-13}	1.07×10^{-11}	3.28
Ta	FDRM	permutation	3.61×10^{-13}	5.23×10^{-12}	2.14
Mo	SNAP	rotation	1.57×10^{-14}	3.28×10^{-13}	2.71×10^{-11}
Mo	SNAP	translation	7.53×10^{-15}	2.25×10^{-13}	1.53×10^{-11}
Mo	SNAP	permutation	1.07×10^{-14}	1.71×10^{-13}	1.34×10^{-11}
Mo	FDRM	rotation	3.69×10^{-14}	6.54×10^{-13}	3.17×10^{-11}
Mo	FDRM	translation	2.23×10^{-14}	3.46×10^{-13}	1.34×10^{-11}
Mo	FDRM	permutation	1.95×10^{-14}	3.14×10^{-13}	1.18×10^{-11}
Nb	SNAP	rotation	9.19×10^{-15}	1.36×10^{-13}	1.08×10^{-11}
Nb	SNAP	translation	3.31×10^{-15}	1.14×10^{-13}	6.66×10^{-12}
Nb	SNAP	permutation	1.29×10^{-15}	7.76×10^{-14}	3.99×10^{-12}
Nb	FDRM	rotation	3.14×10^{-14}	6.58×10^{-13}	1.01×10^{-9}
Nb	FDRM	translation	2.22×10^{-14}	3.37×10^{-13}	5.67×10^{-10}
Nb	FDRM	permutation	2.46×10^{-14}	3.63×10^{-13}	3.93×10^{-10}
Re	SNAP	rotation	8.37×10^{-15}	2.75×10^{-13}	5.92×10^{-12}
Re	SNAP	translation	4.68×10^{-15}	2.68×10^{-13}	3.32×10^{-12}
Re	SNAP	permutation	2.42×10^{-15}	9.37×10^{-14}	2.19×10^{-12}
Re	FDRM	rotation	2.73×10^{-14}	1.59×10^{-12}	1.2×10^{-10}
Re	FDRM	translation	1.76×10^{-14}	7.65×10^{-13}	4.03×10^{-11}
Re	FDRM	permutation	2.27×10^{-14}	8.62×10^{-13}	7.51×10^{-11}

Table 2: Maximum drifts on genuinely excluded configurations: 1000 rotation, 100 translation, and 100 permutation tests per material and method. Forces are aligned to the original frame. Absolute force drift is in eV/Å. Component-relative drift is $d_{F,\text{comp}} = \max_{j,a} |\Delta F_{ja}| / \max(|F_{ja}|, 10^{-12})$; the energy denominator has the same floor. Values above one on Ta arise in almost force-free crystals and must be read alongside their tiny absolute drift. Every individual prediction and transformation is stored.

Material	MD ($\mu\text{s}/\text{atom-step}$)		FDRM evaluation (ms)		Median CV fit (s)	
	SNAP	FDRM	Energy	Energy+force	SNAP	FDRM
Ta	299	858	57.1	109	0.503	0.133
Mo	326	1090	68.5	136	0.569	0.2
Nb	303	881	56.7	112	0.569	0.207
Re	319	1152	79.9	147	0.601	0.258

Table 3: Measured implementation cost. MD values are medians of seven ten-step blocks on 128 atoms. Energy and energy-plus-force columns time FDRM evaluation, including its neighbor search, separately from time integration. Fit columns are medians across the 190 CV training-stage timings per material, using cached geometry and counting pilot learning and descriptor assembly for FDRM. They exclude one-time feature extraction and post-fit reporting; complete timings and all repetitions are logged. CPU inference is single-threaded; training uses one GPU plus the FitSNAP SVD solve.

a compiled, optimized SNAP implementation. FDRM takes 2.87–3.62 times the SNAP time per atom-step across these four probes. The measured ratios describe these actual implementations, not a lower bound on either representation’s cost. A small descriptor vector is not sufficient evidence of a fast potential, and no matched-accuracy speedup is established here.

The code checks analytic forces against finite differences and agreement between direct design-matrix and energy-gradient evaluation. A direct CPU SVD and GPU-QR/FitSNAP-SVD comparison on Ta agrees to 8.86×10^{-11} in the largest prediction difference. The efficient FDRM evaluator agrees with its explicit design-matrix evaluator to 2.05×10^{-12} eV in energy and 5.08×10^{-13} eV/Å in forces on the checked configuration. The isolated-atom tests return zero for FDRM and roundoff-level energy for SNAP. These checks establish internal consistency; they do not validate physical behavior outside the sampled configurations.

6 Scope and implications

The strongest result is a separation of three properties often conflated in descriptor arguments. FDRM is exactly invariant in real arithmetic, preserves its supervised linear pilot, and has fewer fitted head coefficients than quadratic SNAP. Nevertheless it misses full-data accuracy parity on Ta, fails to reach the SNAP full-data threshold with half the data on either Ta aggregation, and regresses on the transfer-material energy comparisons. Favorable isolated cells or the Mo crossing near the end of the sweep do not reverse those outcomes.

The investigation has deliberate limits. A common cutoff and radial grid, no stress fitting, an unregularized scaled least-squares solve, and a single modest tensor construction are not exhaustive choices for either potential family. Singular-value truncation after column scaling is an implicit numerical selection rule, even without a ridge penalty. The learned mode matrices carry additional fitted degrees of freedom beyond the head count. The supplied Ta inventory and archived reference fit differ from the published summary, and random configuration splits of adaptive-learning trajectories can remain optimistic despite the duplicate control. The short timing trajectories do not establish reliable phonons, defects, melting behavior, elastic response, or long-term MD stability. We make no claims about these properties.

These limits also specify the next scientific test. Restoring informative volume derivatives through stress observations, controlling lower and higher body orders at comparable capacity, and selecting numerical regularization entirely within the included portion could distinguish coverage from expressivity. Those extensions would constitute new experiments. The present completed study supports a narrower conclusion: supervised spectral compression can learn useful invariant radial channels, but the particular response preserved by a pilot is not a certificate of data efficiency for its quadratic extension.

Reproducibility and artifacts. Code, fitted models, configuration indices, per-run settings, timestamps, and predictions accompany the paper. `logs/all_cells.csv` retains every draw’s included/excluded metrics; `logs/summary_cells.csv` gives their means and sample standard deviations. Published values are identified in `logs/paper_reference.md`; code inspection and execution attempts are documented separately. Appendix D and `README.md` give reconstruction instructions.

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A Additional paired measurements

Material	Method	80% energy MAE (eV/atom)	80% force MAE (eV/Å)
Ta	Quadratic SNAP	3.568 ± 7.09	0.0799 ± 0.0173
Ta	FDRM	1.207 ± 2.14	0.06138 ± 0.0154
Ta	Fixed radial modes	8.885 ± 23.6	0.148 ± 0.209
Ta	Random radial modes	0.6541 ± 1.42	0.06041 ± 0.0129
Mo	Quadratic SNAP	0.06377 ± 0.0181	0.6947 ± 0.0549
Mo	FDRM	0.06624 ± 0.0176	0.6594 ± 0.0338
Mo	Fixed radial modes	0.06634 ± 0.0186	0.7485 ± 0.0371
Mo	Random radial modes	0.06615 ± 0.0184	0.6896 ± 0.0301
Nb	Quadratic SNAP	0.005252 ± 0.00321	0.1336 ± 0.00844
Nb	FDRM	0.01609 ± 0.00469	0.4975 ± 0.0219
Nb	Fixed radial modes	0.01443 ± 0.00292	0.532 ± 0.0231
Nb	Random radial modes	0.02011 ± 0.0035	0.7124 ± 0.0632
Re	Quadratic SNAP	0.009813 ± 0.00116	0.3238 ± 0.00781
Re	FDRM	0.04993 ± 0.0046	1.273 ± 0.204
Re	Fixed radial modes	0.05079 ± 0.00357	1.481 ± 0.222
Re	Random radial modes	0.05252 ± 0.00555	1.436 ± 0.214

Table 4: Excluded equal-group means $\pm$ sample standard deviations at the 80% budget, across all ten draws. Fixed and random controls retain the same 40 descriptor channels and 860-coefficient head as FDRM. Random orthogonal matrices use the run seed and are independent of labels. Both controls are fitted on the same included configurations as the paired main models. All other tested budgets are included in the machine-readable tables.

A.1 Restoring the supplied short-range reference

All rows below use the same Ta configurations, mode-learning rule, weights, and paired draws as the main experiment, with the supplied ZBL reference subtracted before fitting and added at prediction. The full-data line is a single fit; all other lines are excluded means $\pm$ sample standard deviations over ten draws. Energies are in eV/atom.

Budget	Method	All-group energy	Low-symmetry energy
20%	SNAP	$7.078 \times 10^4 \pm 8.635 \times 10^4$	38.78 ± 25.85
20%	FDRM	1465 ± 3774	0.1149 ± 0.2634
50%	SNAP	12.47 ± 11.94	0.01965 ± 0.01816
50%	FDRM	18.89 ± 18.8	0.009437 ± 0.008047
80%	SNAP	1.923 ± 2.346	0.01771 ± 0.01957
80%	FDRM	1.069 ± 1.65	0.0156 ± 0.02088
95%	SNAP	0.4548 ± 0.6259	0.01936 ± 0.03133
95%	FDRM	6.138 ± 18.75	0.01081 ± 0.01954
100% fit	SNAP	0.0003143	0.0005475
100% fit	FDRM	0.0004944	0.0007814

A.2 Coordinate-cluster split control

Coordinates and cell entries are rounded to eight decimal places before hashing; atom ordering is retained. Eighty percent of global hash clusters are selected with seed $700001 + d$, where $d = 0, \dots, 9$. The split is not stratified by source group. All labels for every member of a cluster stay together. The table reports excluded equal-group means and sample standard deviations.

Material	Method	Energy MAE (eV/atom)	Force MAE (eV/Å)
Mo	SNAP	0.06474 ± 0.03403	0.7228 ± 0.04707
Mo	FDRM	0.06827 ± 0.03396	0.6678 ± 0.03192
Nb	SNAP	0.006519 ± 0.003598	0.1362 ± 0.01435
Nb	FDRM	0.01778 ± 0.006241	0.4839 ± 0.01544
Re	SNAP	0.009791 ± 0.001155	0.3216 ± 0.009981
Re	FDRM	0.04884 ± 0.005339	1.262 ± 0.1925

The actual Mo configuration fraction ranges from 79.85% to 80.14%. The actual Nb configuration fraction ranges from 79.64% to 80.19%. The actual Re configuration fraction ranges from 79.72% to 80.15%.

A.3 Pilot retention and derivative checks

The spectral map identity is checked on all energy and force design rows. The squared residuals below use the energy/force weights of Eq. (7). They combine different physical units through that prescribed weighting and should be compared within a material, not interpreted as a physical energy.

Material	Pilot weighted SSE	Head weighted SSE	Max. embedding discrepancy
Ta	293.4	35.66	1.137×10^{-12}
Mo	3.034×10^6	2.135×10^6	2.572×10^{-12}
Nb	2.258×10^6	2.707×10^5	3.05×10^{-12}
Re	5.372×10^6	2.884×10^6	1.075×10^{-12}

Central finite differences of the trained energy, with a 10^{-5} Å displacement on the documented excluded configurations, differ from analytic force components by at most 7.971×10^{-8} eV/Å. This is a numerical spot check, not a bound over configuration space.

B Full-data group errors

The following are fitting errors, not held-out predictions. Energy units are eV/atom and force units are eV/Å. Force components are averaged within each group. Every sub-100% included and excluded cell, including all ten draws, is separately available in the accompanying CSV files.

Material	Group	SNAP δE	FDRM δE	SNAP δF	FDRM δF
Ta	Displaced_A15	0.0001434	0.0001664	0.0251	0.03574
Ta	Displaced_BCC	0.0001877	0.0004485	0.04252	0.05944
Ta	Displaced_FCC	0.0001765	0.0002332	0.02151	0.02833
Ta	Elastic_BCC	0.0001195	0.0001483	2.361×10^{-5}	2.226×10^{-5}
Ta	Elastic_FCC	0.000236	0.00025	0.000107	0.0001028
Ta	GSF_110	0.0003758	0.0004543	0.007357	0.01183
Ta	GSF_112	0.0005287	0.000979	0.01498	0.02214
Ta	Liquid	0.0002045	0.0002541	0.02605	0.1063
Ta	Surface	0.001115	0.001512	0.00852	0.01311
Ta	Volume_A15	0.0002128	0.0003254	1.104×10^{-13}	2.442×10^{-13}
Ta	Volume_BCC	0.000371	0.000748	4.281×10^{-13}	8.605×10^{-13}
Ta	Volume_FCC	0.0001738	0.0003994	1.171×10^{-13}	1.725×10^{-13}
Mo	iter0_elastic	0.01029	0.01472	0.461	0.5037
Mo	iter0_trajectory	0.006767	0.009089	0.5106	0.6406
Mo	iter0_vacancies	0.01115	0.01614	0.6257	0.6333
Mo	iter1_elastic	0.004376	0.007301	0.2717	0.3692
Mo	iter1_trajectory	0.006158	0.01751	0.5141	0.6805
Mo	iter1_vacancies	0.008444	0.0268	0.3592	0.2577
Mo	iter2_elastic	0.1561	0.1524	0.8347	0.7273
Mo	iter2_trajectory	0.2975	0.2903	1.163	0.9321
Mo	iter2_vacancies	0.02399	0.02156	0.8893	0.7248
Nb	iter0_elastic	0.00238	0.009652	0.1265	0.5906
Nb	iter0_trajectory	0.003384	0.01441	0.1187	0.4652
Nb	iter0_vacancies	0.001554	0.01058	0.1053	0.339
Nb	iter1_elastic	0.01304	0.01796	0.09933	0.363
Nb	iter1_trajectory	0.002246	0.009087	0.1012	0.473
Nb	iter1_vacancies	0.001372	0.007052	0.09812	0.3721
Nb	iter2_elastic	0.008039	0.01293	0.1006	0.3374
Nb	iter2_trajectory	0.001869	0.01183	0.101	0.4556
Nb	iter2_vacancies	0.001245	0.006866	0.09816	0.3722
Nb	iter3_elastic	0.003158	0.01691	0.1937	0.7156
Nb	iter3_trajectory	0.004213	0.01889	0.2048	0.7477
Nb	iter3_vacancies	0.004886	0.01859	0.1576	0.557
Re	iter0_elastic	0.004357	0.01933	0.2985	1.115
Re	iter0_trajectory	0.007012	0.04054	0.2921	1.288
Re	iter0_vacancies	0.007055	0.03907	0.2328	0.9235
Re	iter1_elastic	0.009126	0.04241	0.3348	1.149
Re	iter1_trajectory	0.009843	0.04129	0.3661	1.387
Re	iter1_vacancies	0.01174	0.06402	0.323	1.353
Re	iter2_elastic	0.008771	0.04144	0.3453	1.173
Re	iter2_trajectory	0.01019	0.04949	0.3714	1.384
Re	iter2_vacancies	0.01211	0.0582	0.3159	1.229

C Per-group force and transfer curves

All figures use the same conventions as the main learning curves: solid lines and bands indicate excluded means and sample standard deviations over ten draws, disconnected hollow circles at 100% indicate full-data fitting errors, and dotted lines identify the refitted SNAP full-data reference. Bands are clipped only for rendering on logarithmic axes; raw values are retained. Energy curves for all twelve Ta groups appear in the main text.

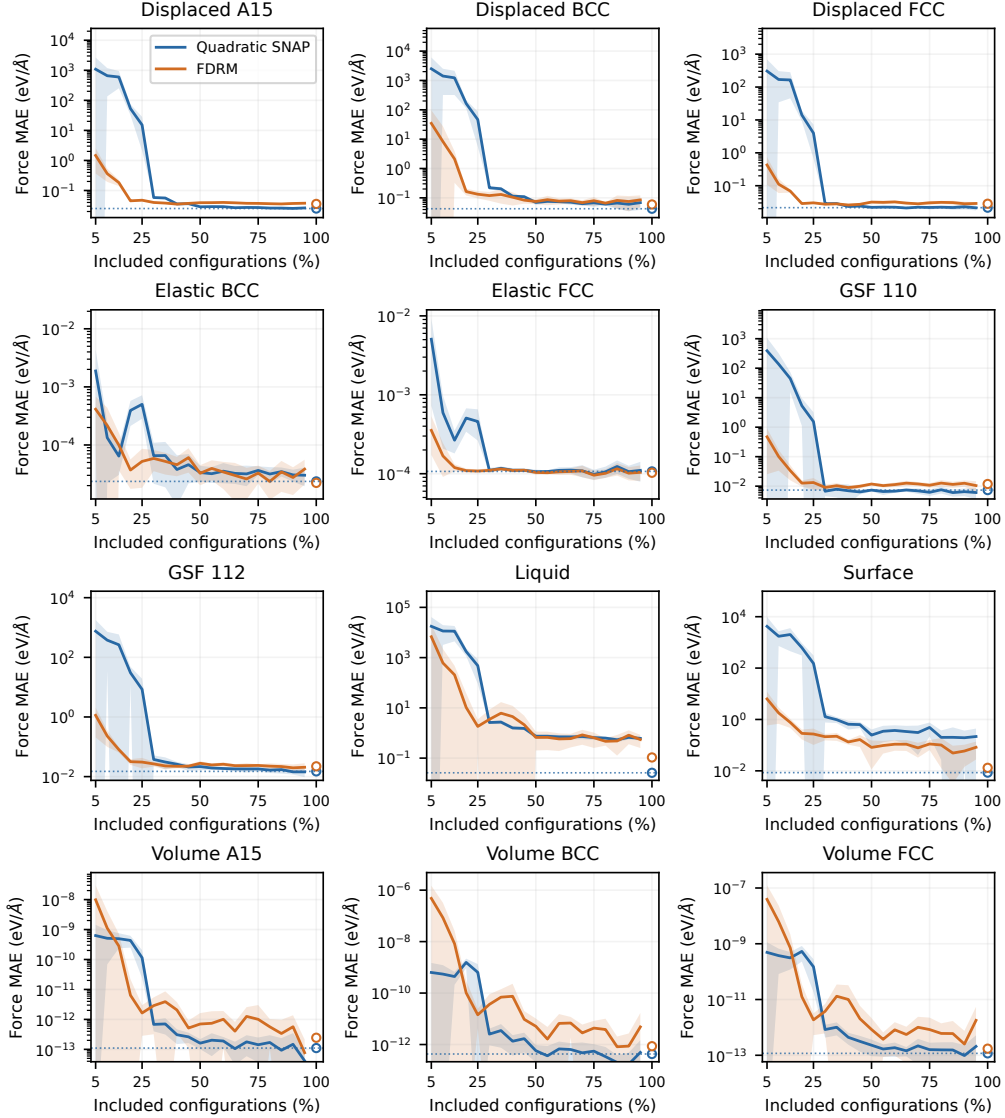


Figure 5: All twelve Ta groups: force MAE. Very small values for volume scans reflect symmetry-forced vanishing atomic forces; they do not imply accurate hydrostatic energy derivatives.

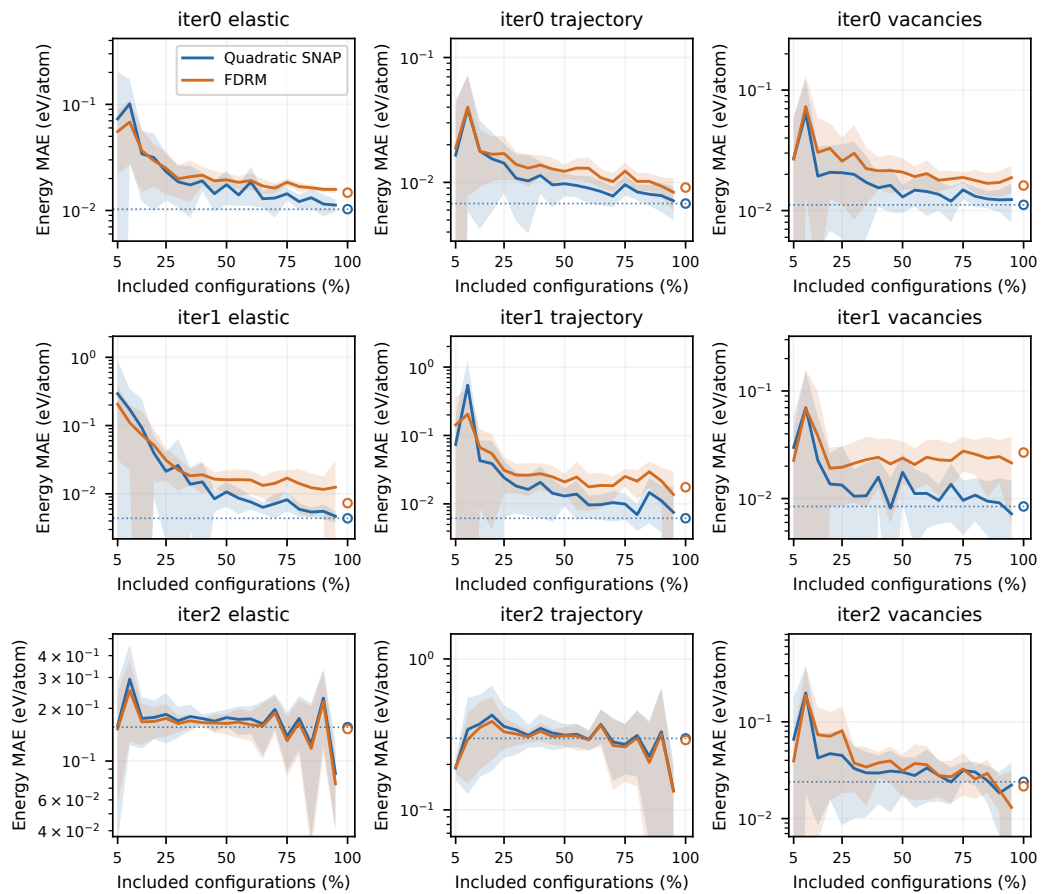


Figure 6: Mo energy MAE in every iteration/perturbation stratum.

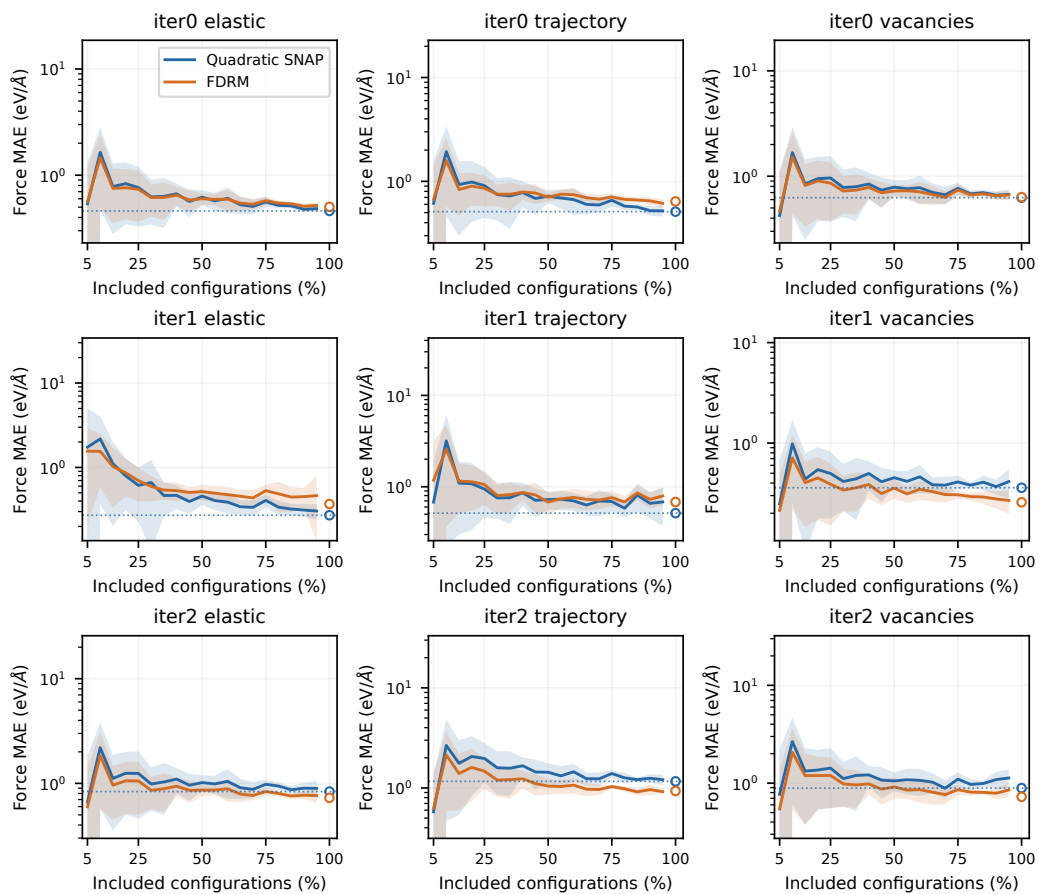


Figure 7: Mo force MAE in the same strata.

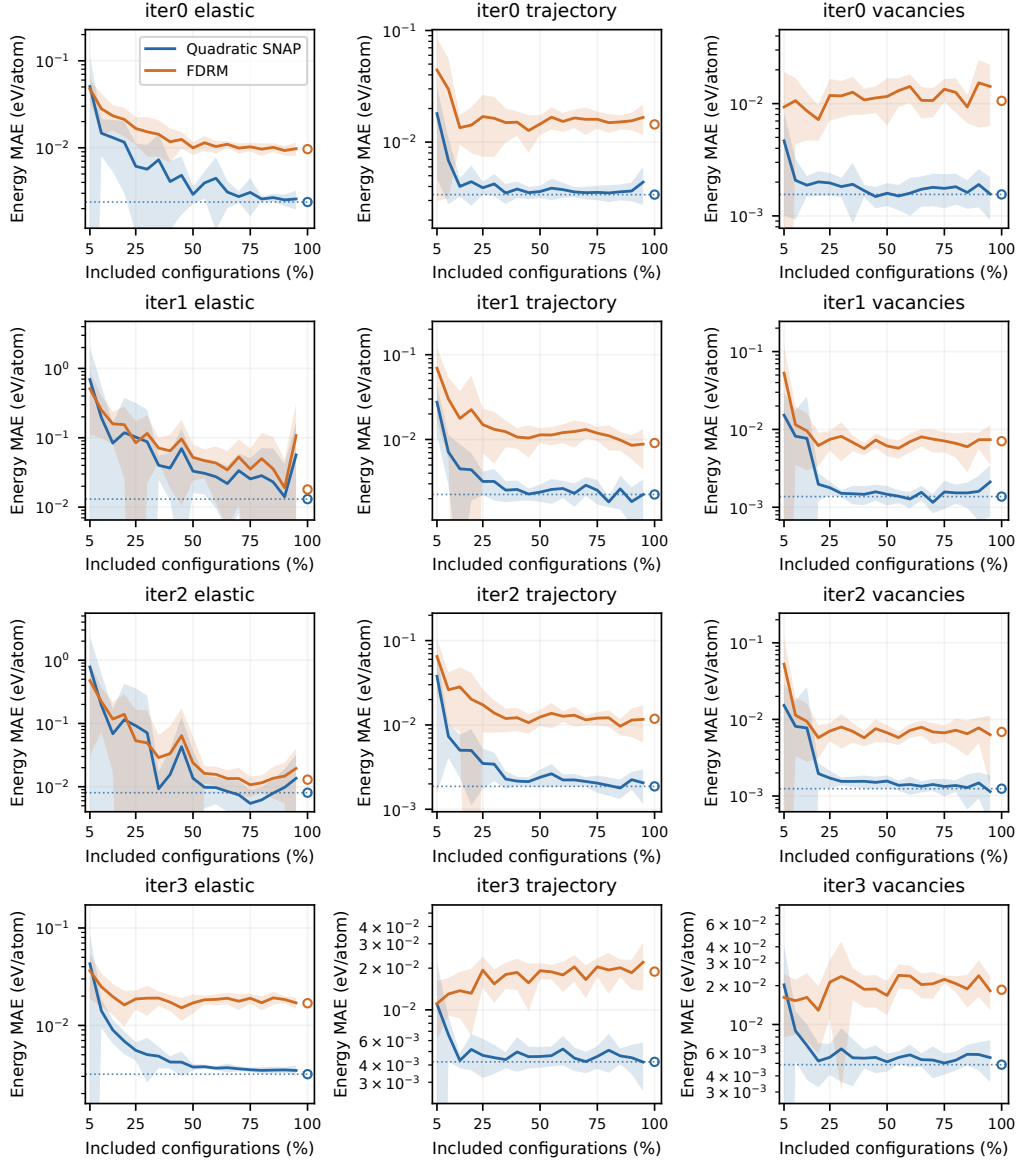


Figure 8: Nb energy MAE in every iteration/perturbation stratum.

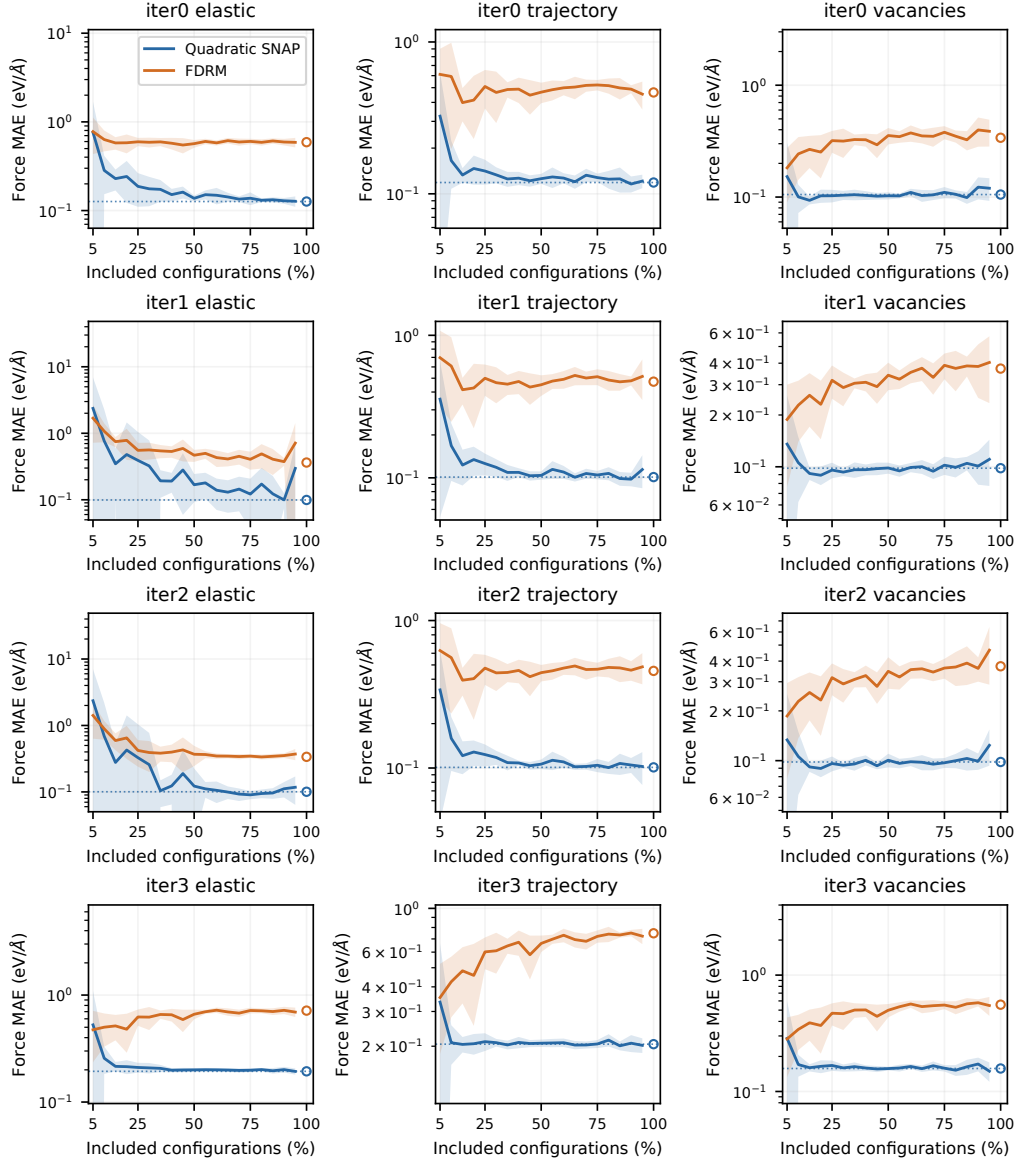


Figure 9: Nb force MAE in the same strata.

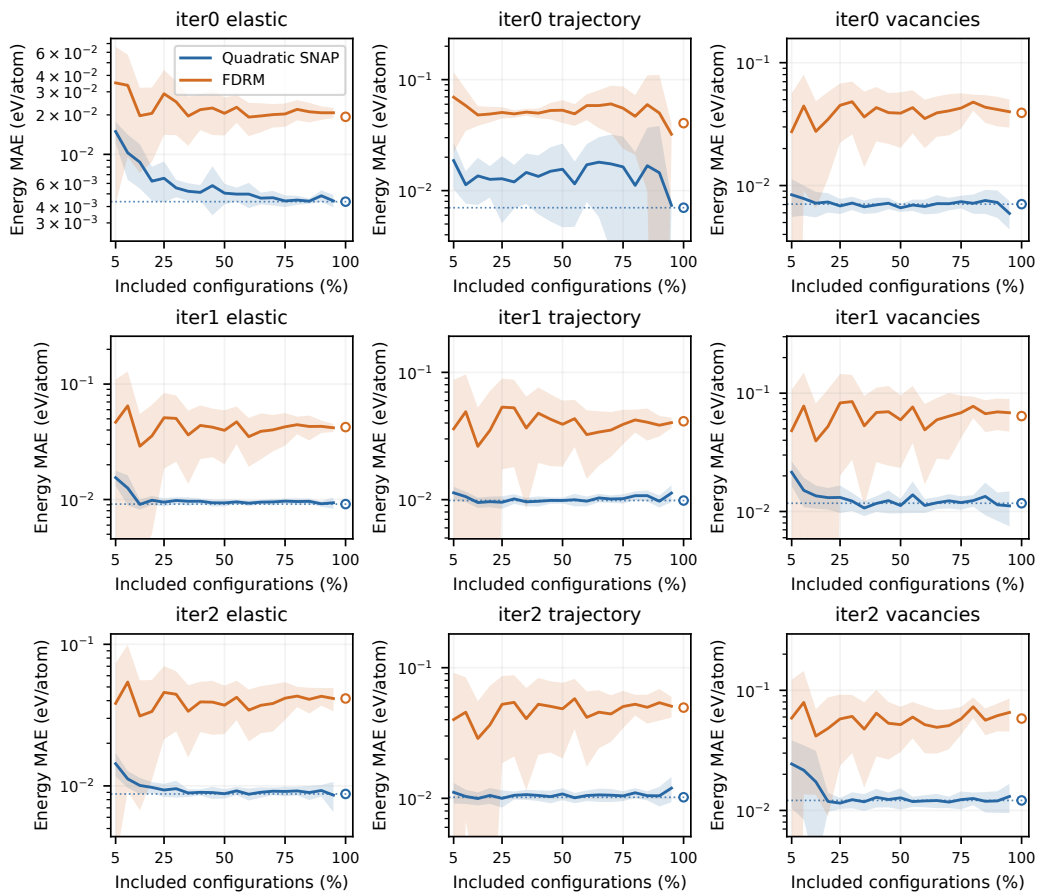


Figure 10: Re energy MAE in every iteration/perturbation stratum.

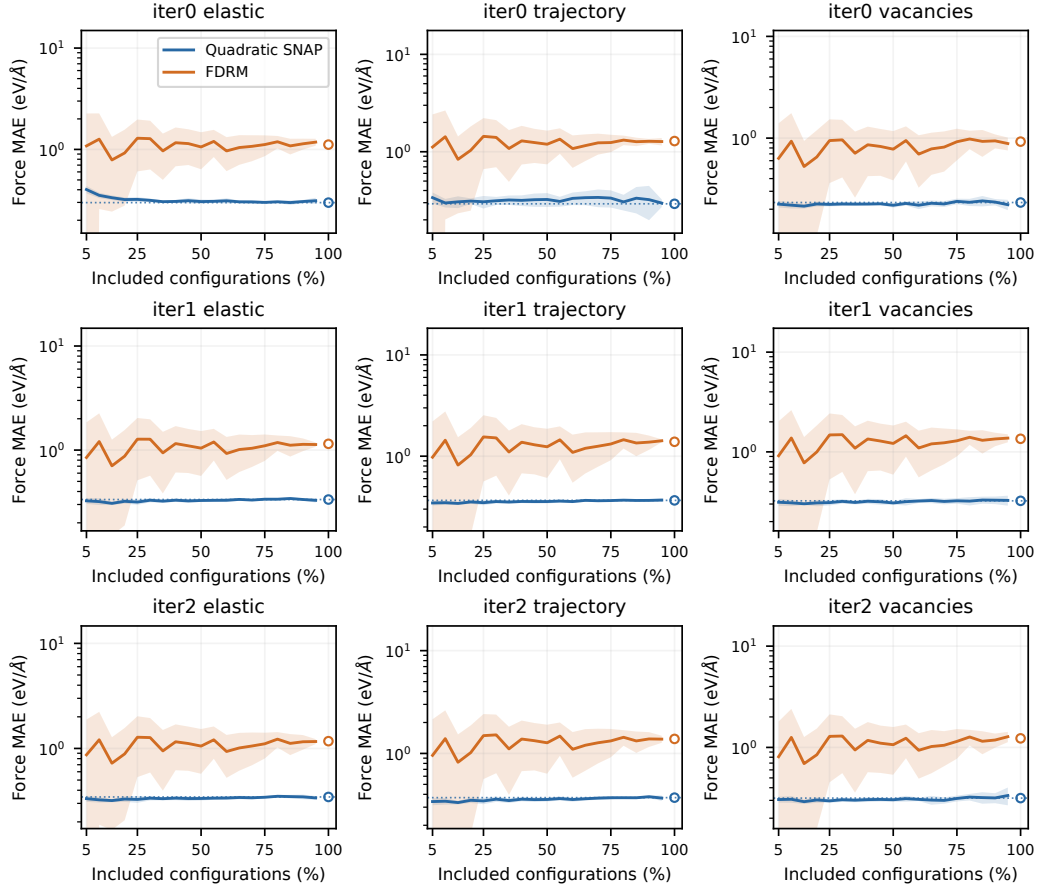


Figure 11: Re force MAE in the same strata.

D Artifact inventory and reconstruction

Material	Configurations	Atoms	Force components	Strata
Ta	363	4224	12672	12
Mo	3727	66732	200196	9
Nb	3246	56386	169158	12
Re	5420	112134	336402	9

The primary experiment contains 1528 completed fitted heads: two methods, four materials, nineteen fractions with ten draws, and one full-data fit per material/method. The fixed and random controls add 248 heads. The Ta reference control adds 82 and the duplicate-cluster control adds 60. Intermediate pilot fits are logged as training stages, not counted as additional competing methods. Numerical spot checks and timing measurements are separately identified. Ta group sizes, in the order of Fig. 2, are 9, 9, 9, 100, 100, 22, 22, 3, 7, 30, 21, and 31. This discreteness is why nominal and effective budgets differ. The transfer data retain every available frame from the supplied arrays, including exact-coordinate duplicates. The MD probe uses conventional BCC lattice constants 3.30, 3.15, and 3.30 Å for Ta, Mo, and Nb. Re uses an orthorhombic HCP representation with $a = 2.76$ and $c = 4.45$ Å. These are chosen timing geometries, not DFT-derived structural predictions. Seeds for initial velocities are 9137; the independent symmetry-audit selection seed is 480 and transformation seeds are 9371 + configuration id. README.md gives reproduction commands; logs/ records full data and model provenance.